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SYSC 3501 Communication Theory

Lab report / Sign-off sheet

Lab no: #2

Lab Title: Time and Frequency Responses of LTI Systems, Butterworth Filters

Lab Section: L40

Name and signature of TA: 

Lab Date: February 6th, 2025

Late Attendance: No (When checked, grade will be reduced by 20%)

Lab Grade: (To be Filled by TAs)

Date Completed: February 12th, 2025

Group Members:

#	Name	Student #
1	JABER-UL HUDA	
2		

TA's should fill out the information in the boxes. Each student should present this sheet to the Lab TA when leaving the lab. This sheet should be attached to the lab report. Lab reports with no sign-off page will receive a mark of 0.

NAME: JABER-UL HUDA, ID: 101137524

Pre-Lab

Q1. Give the theoretical time-domain steady-state expressions of the output of the filter when the input is a sine wave of frequency 1Hz, and low-pass Butterworth filters of cut-off frequencies 100Hz, 2Hz, 1Hz are used.

A: When a sine wave of frequency 1 Hz is passed through a low-pass Butterworth filter, the output is a sine wave of the same frequency but with reduced amplitude and a phase shift, depending on the filter's cutoff frequency. When processed through low-pass Butterworth filters with cutoff frequencies of 100Hz, 2Hz, and 1Hz, the steady-state output expression can be formulated as:

$$y(t) = |H(1\text{Hz})| \times A \times \sin 2\pi t + \phi$$

Where:

- A represents the amplitude of the input sine wave.
- $|H(1\text{ Hz})|$ denotes the magnitude of the filter's frequency response at 1Hz.
- ϕ is the phase shift introduced by the filter at 1 Hz.

For a filter with a 100Hz cutoff frequency, the magnitude of the transfer function at 1Hz ($|H(1\text{ Hz})|$) will be approximately 1, as the input frequency is significantly below the cutoff threshold. In contrast, for a filter with a 1Hz cutoff, $|H(1\text{ Hz})|$ will be at the -3dB point, indicating a substantial reduction in signal strength. For a 2Hz cutoff, $|H(1\text{ Hz})|$ will lie between 0 and 1, suggesting partial attenuation due to the proximity of the input frequency to the filter's cutoff point.

Q2. Design a Matlab program that would plot $|H(f)|$ theoretical for a 8th order Butterworth filter with its 3-dB cutoff frequency as parameter, save this program on a floppy and bring it to the lab. We may use that program to answer Q1.4 of the lab manual.

A: Design a Matlab program that would plot $|H(f)|$ theoretical for a 8th order Butterworth filter with its 3-dB cutoff frequency as parameter, save this program on a floppy and bring it to the lab. We may use that program to answer Q1.4 of the lab manual. `function plot_butterworth(fc) % fc is the cutoff frequency [b, a] = butter(8, fc, 'low', 's'); % Generate filter coefficients h = freqz(b, a, logspace(-1, 2, 1000)); % Frequency response of filter semilogx(linspace(-1, 2, 1000), 20*log10(abs(h))); % Plot in dB scale grid on; title('8th Order Butterworth Filter Frequency Response'); xlabel('Frequency (Hz)'); ylabel('Magnitude (dB)'); xlim([0.1 100]); % Set x-axis limits to show from 0.1Hz to 100Hz`

Q3. Write the Fourier Series expansion of a periodic square wave of frequency 1Hz. Deduce the theoretical time-domain steady-state expressions of the output of the filter when the input is a periodic square wave of frequency 1Hz and low-pass Butterworth filters of cut-off frequencies 100Hz, 4Hz, 2Hz and 1Hz are used.

A: A periodic square wave of frequency $f_0 = 1\text{ Hz}$ with an amplitude of A can be expressed as a Fourier series:

$$w(t) = \frac{4A}{\pi} \sum_{n=1}^{\infty} \frac{1}{n} \sin(2\pi f_0 t)$$

For the case when $f_c = 100\text{ Hz}$ Since 100 Hz is much greater than all significant harmonics of the 1Hz square wave (1 Hz, 3 Hz, 5 Hz, etc.), the filter has minimal effect. The output is nearly identical to the original square wave.

For the case when $f_c = 4\text{ Hz}$ This filter significantly attenuates harmonics beyond 4 Hz. The harmonics at 5 Hz, 7 Hz, etc., are strongly suppressed. The resulting waveform is smoother, resembling a fundamental sine wave plus a small contribution from the third harmonic

For the case when $f_c = 2$ Hz the harmonics at 3 Hz is significantly attenuated, leaving only the fundamental component. This means the output is approximately a pure sine wave of frequency 1 Hz.

For the case when $f_c = 1$ Hz the fundamental component at 1 Hz is partially attenuated, but higher-order harmonics are completely suppressed. The system behaves like a first-order low-pass filter, and the output is a strongly attenuated sine wave. This shows that as the cutoff frequency decreases, the output wave increasingly resembles a smoothed, attenuated sine wave

Q4. Using the Fourier Series expansion of a periodic square wave of frequency 1Hz, deduce its theoretical amplitude spectrum.

A: A periodic square wave of frequency $f_0 = 1$ Hz with an amplitude of A can be expressed as a Fourier series:

$$w(t) = \frac{4A}{\pi} \sum_n^{\infty} \frac{1}{n} \sin(2\pi f_0 t)$$

A square wave is an odd function, meaning it can be represented by a Fourier sine series with only odd harmonics. The amplitude spectrum shows the magnitude of each harmonic component in the frequency domain. From the Fourier series, the amplitudes of the sine components are:

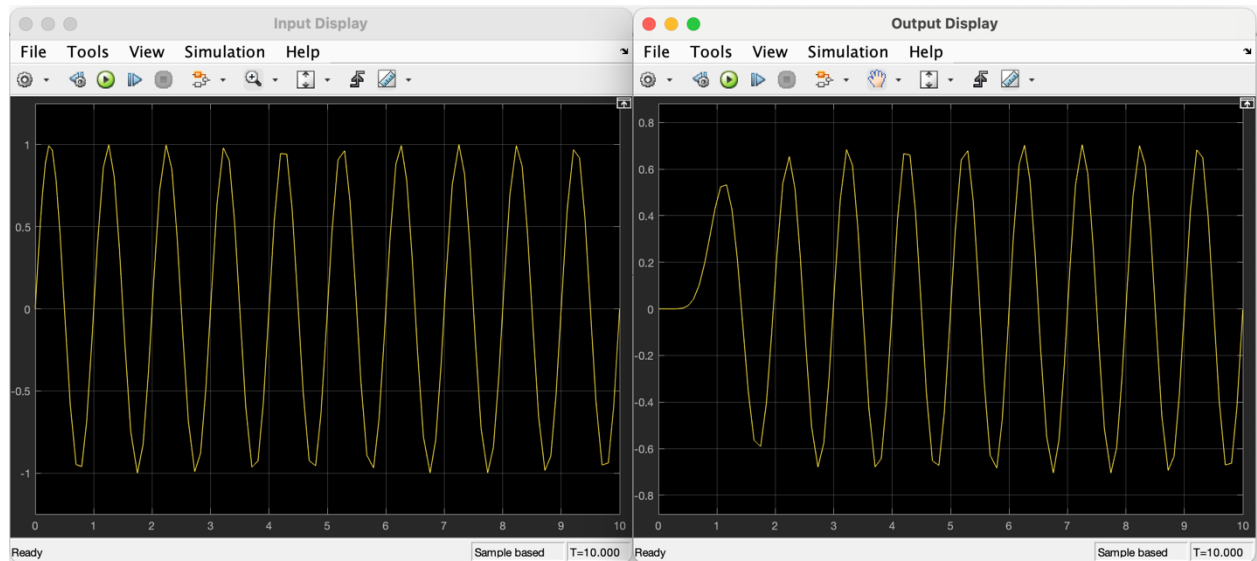
$$|C_n| = \frac{4A}{n\pi}, \text{ for odd } n$$

Thus, the theoretical amplitude spectrum consists of discrete spikes at odd harmonics $f_n = nf_0 = n$ Hz (where n is odd), with amplitudes given by:

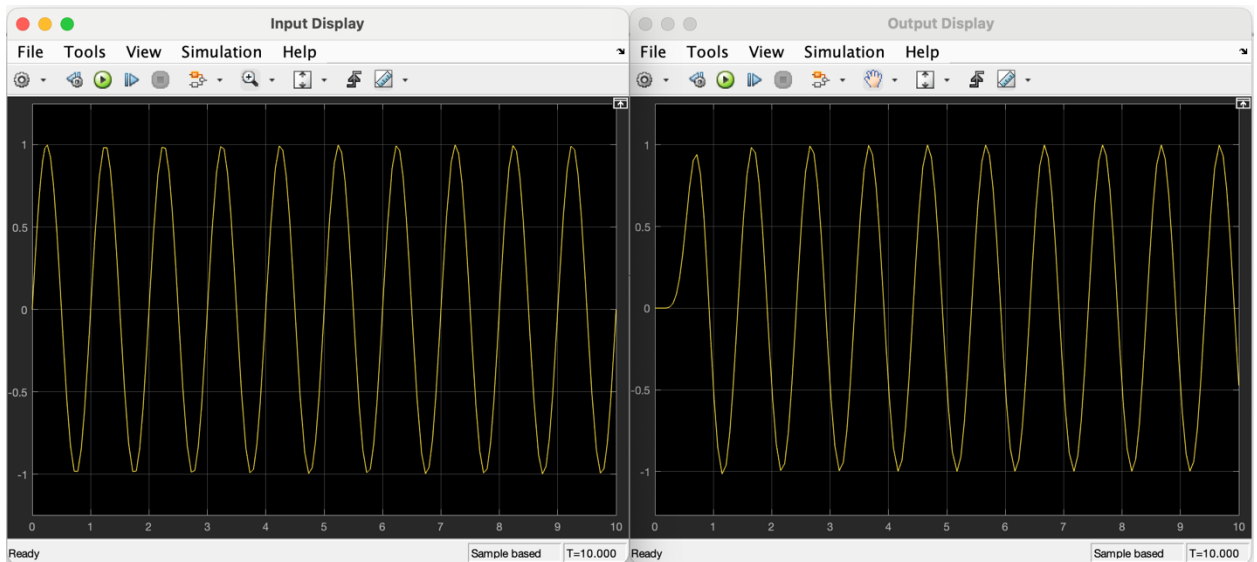
$$|C_n| = \frac{4A}{n\pi}, \text{ for odd } n$$

Part 1

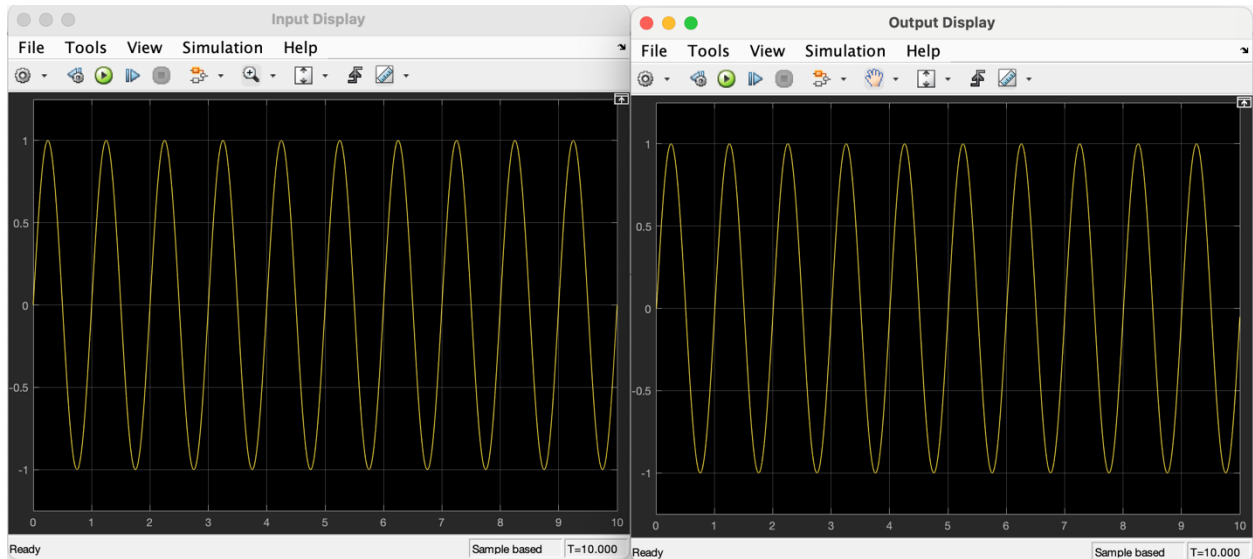
Q 1.1:



Cut-off frequency 1 Hz



Cut-off frequency 2 Hz



Cut-off frequency 100 Hz

Q 1.2:

For a filter with a 100Hz cutoff frequency, the output signal's amplitude remains constant at 1, indicating no attenuation for a 1 Hz sine wave passing through it. This aligns with the expected behavior for a cutoff frequency significantly higher than the signal frequency. Additionally, the delay is observed to be 0, suggesting no phase shift for the 1 Hz frequency component due to the filter.

When the cutoff frequency is set to 2 Hz, the amplitude of the output signal remains constant at 1, indicating that the 1 Hz sine wave falls within the passband. However, a slight phase shift of 0.3 degrees is observed, which could reflect the filter's phase response near the cutoff frequency. This phase shift may potentially affect lower frequencies.

For a cutoff frequency of 1 Hz, the amplitude of the output signal is reduced to 0.7, indicating that the 1 Hz sine wave is approaching the edge of the cutoff frequency, leading to signal attenuation by the filter.

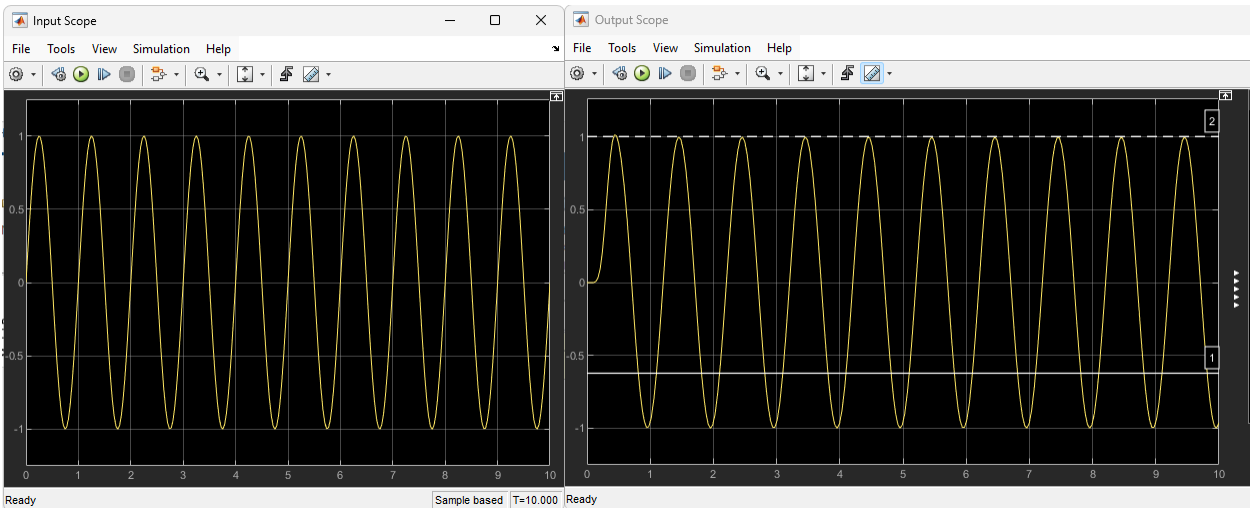
Furthermore, a significant phase delay of 0.5 is observed, suggesting that the 1 Hz component is either affected by the filter's transition band or is very close to the cutoff frequency, resulting in a larger phase delay.

Frequency	Amplitude	Delay
100 Hz	1	0
2 Hz	1	0.3
1 Hz	0.706	0.5

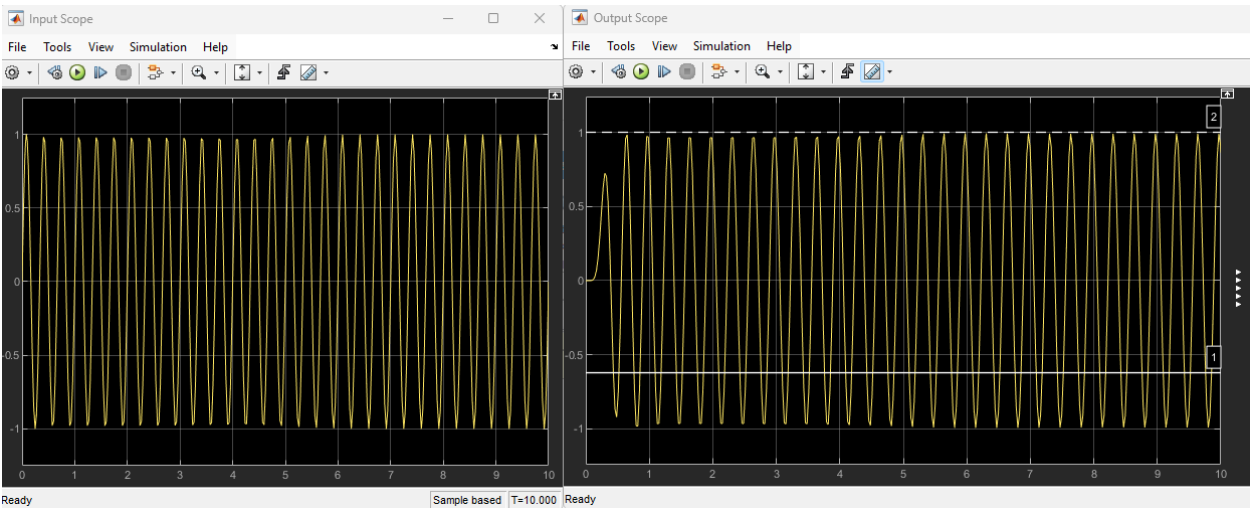
Setting the cut-off frequency of the filter to 4 Hz. Varying the frequency of the sine wave input from 1 Hz to 10 Hz and measure the amplitude of the output sine wave.

Frequency	Amplitude	DB
1	1	0
3	1	0
3.5	0.94	0
4.5	0.351	-4.55
5	0.158	-8.01
7	0.012	-19.21
10	0.0006314	-32

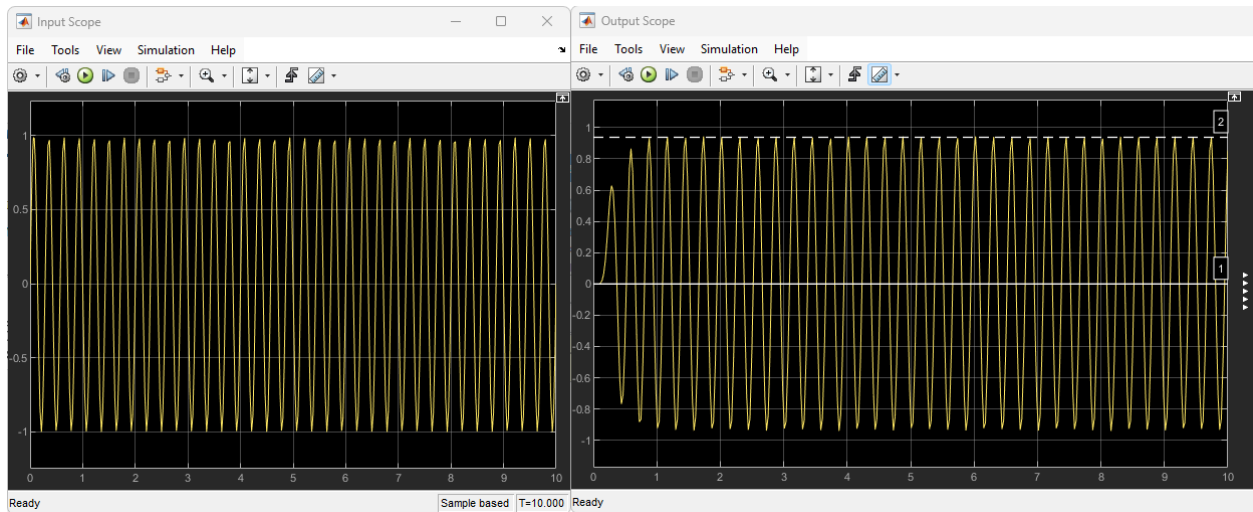
The picture of the waveforms of each frequency from 1Hz to 100 Hz when the cut-off is set to 4 Hz:



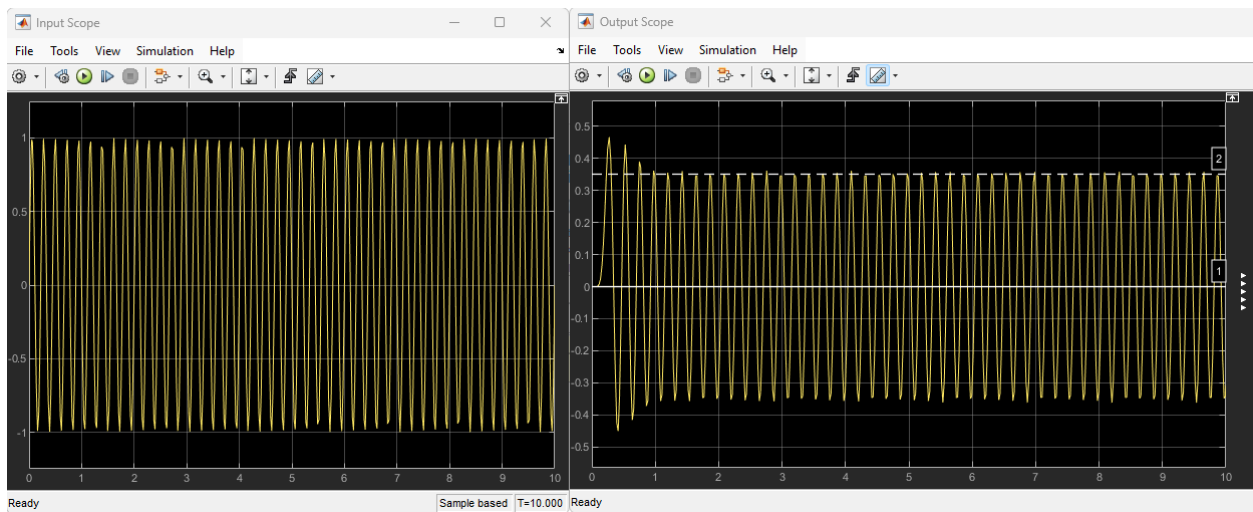
1Hz input



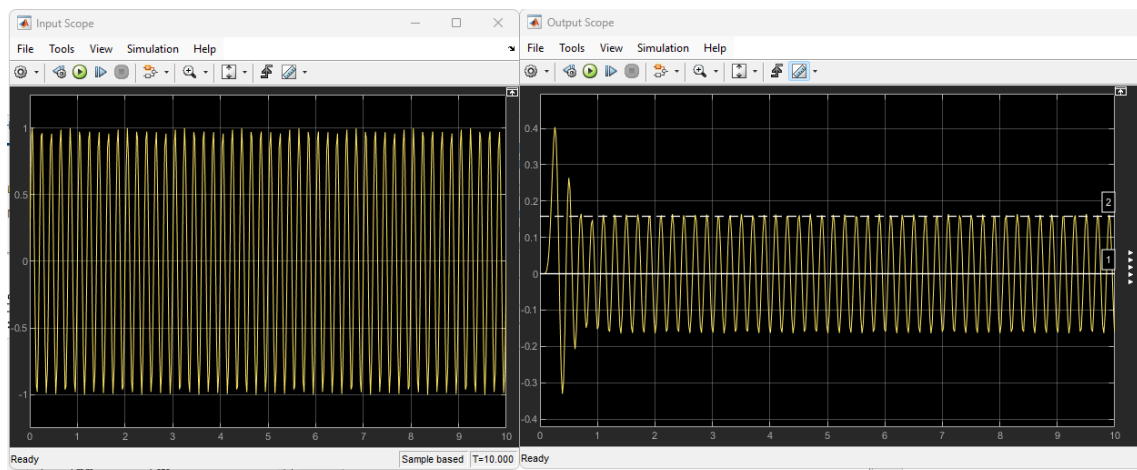
3Hz input



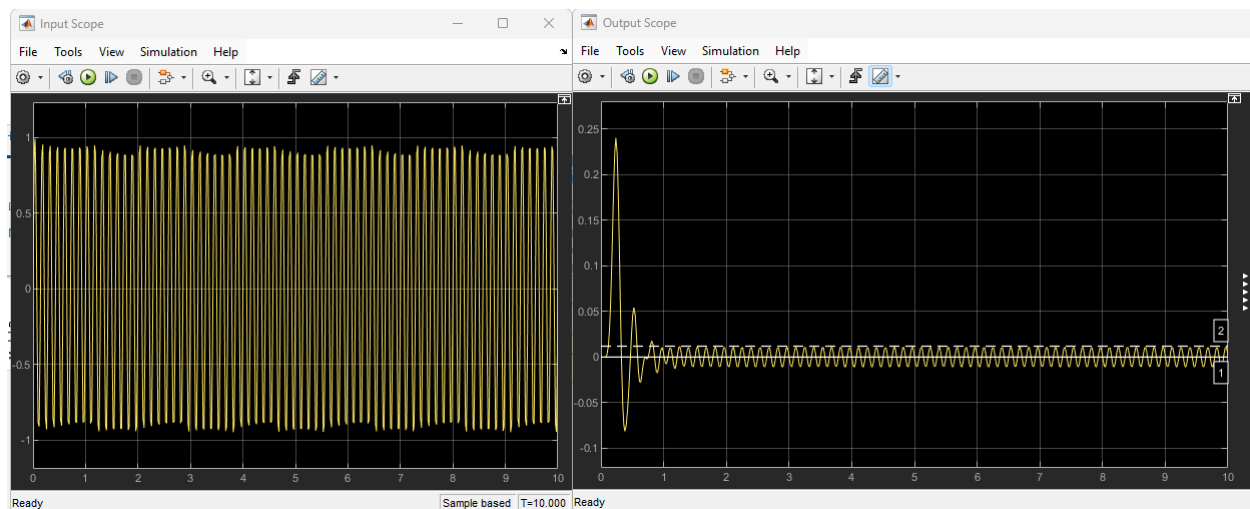
3.5Hz input



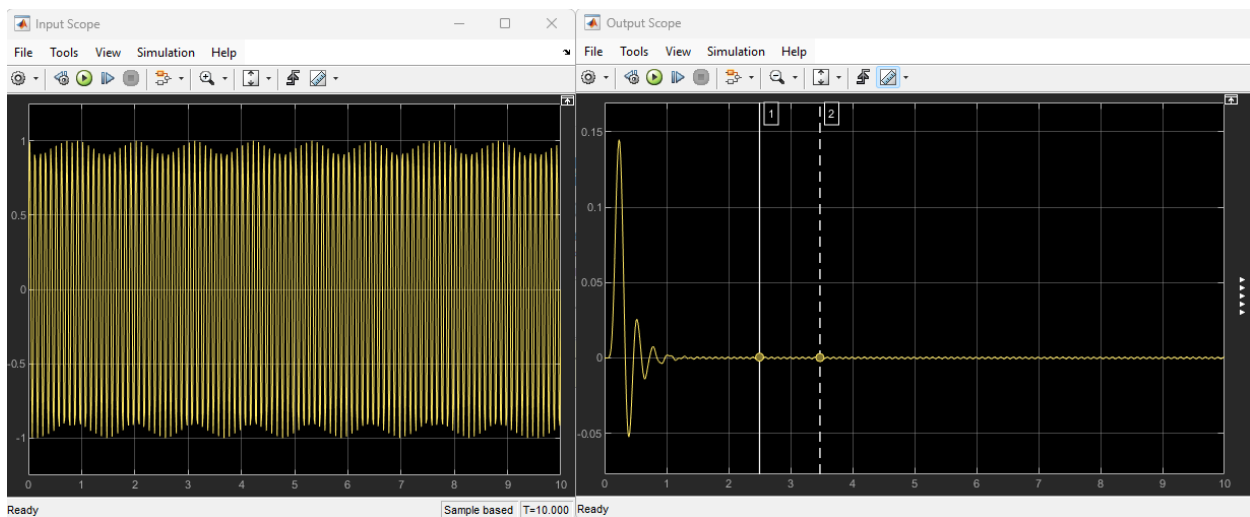
4.5Hz input



5Hz input



7Hz input



10Hz input

Q 1.3:

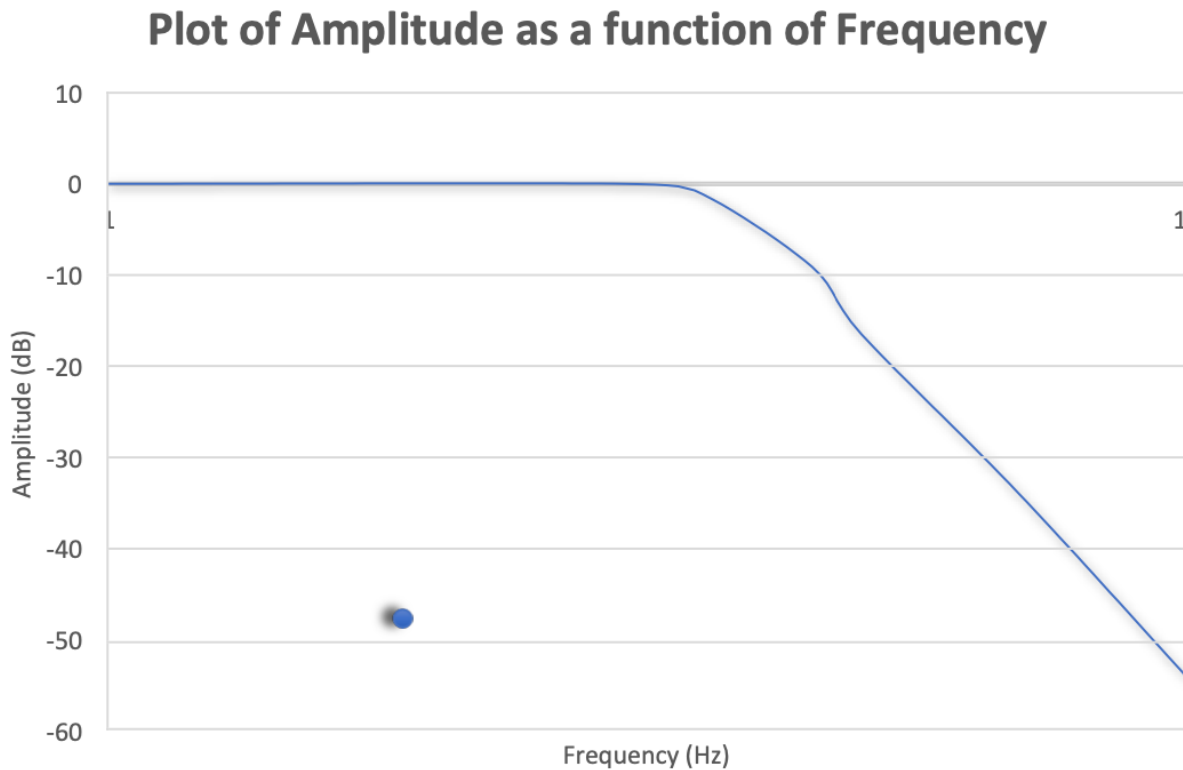


Figure: 1

The graph in Figure 1 shows the relationship between the frequency and amplitude of the signal. As the frequency is increased, the amplitude remains at 0 dB at the early stage, however, after the frequency is increased to 3.5 Hz, the amplitude keeps decreasing. As the frequency is increased from 3.5 to 10 Hz, the amplitude continues to have a sharp drop exponentially.

Q 1.4:

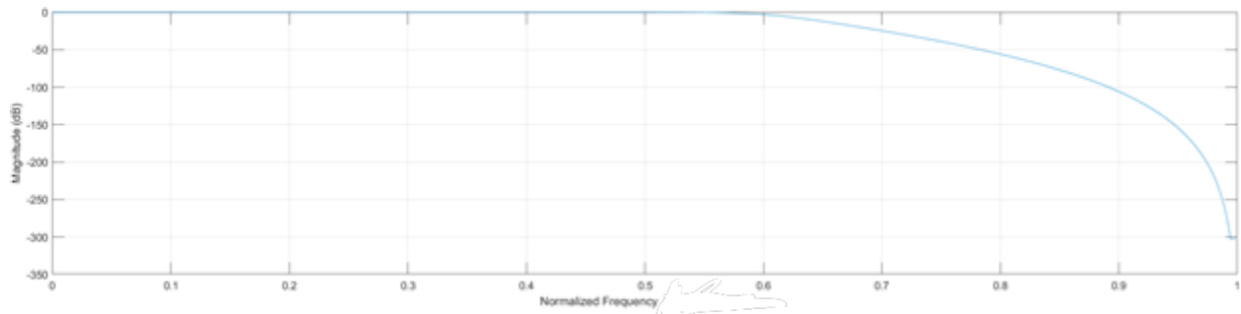


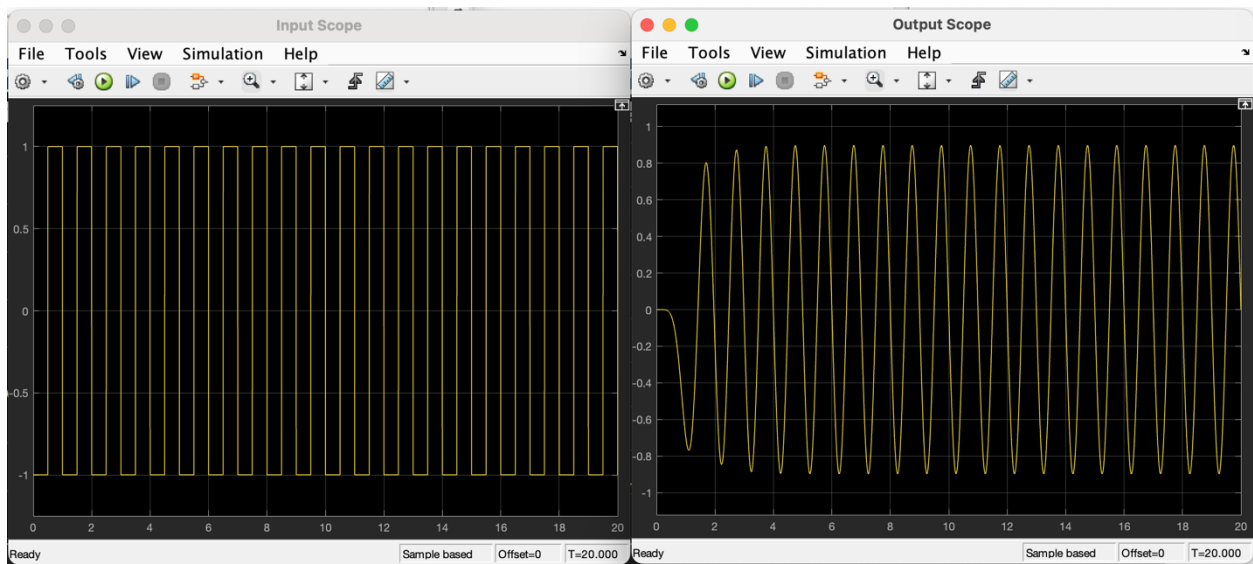
Figure 2

The theoretical graph as shown in Figure 2 also shows that as the frequency is increased, the amplitude decreases exponentially, which represents the same concept as shown in Figure 1 for the experimental plot.

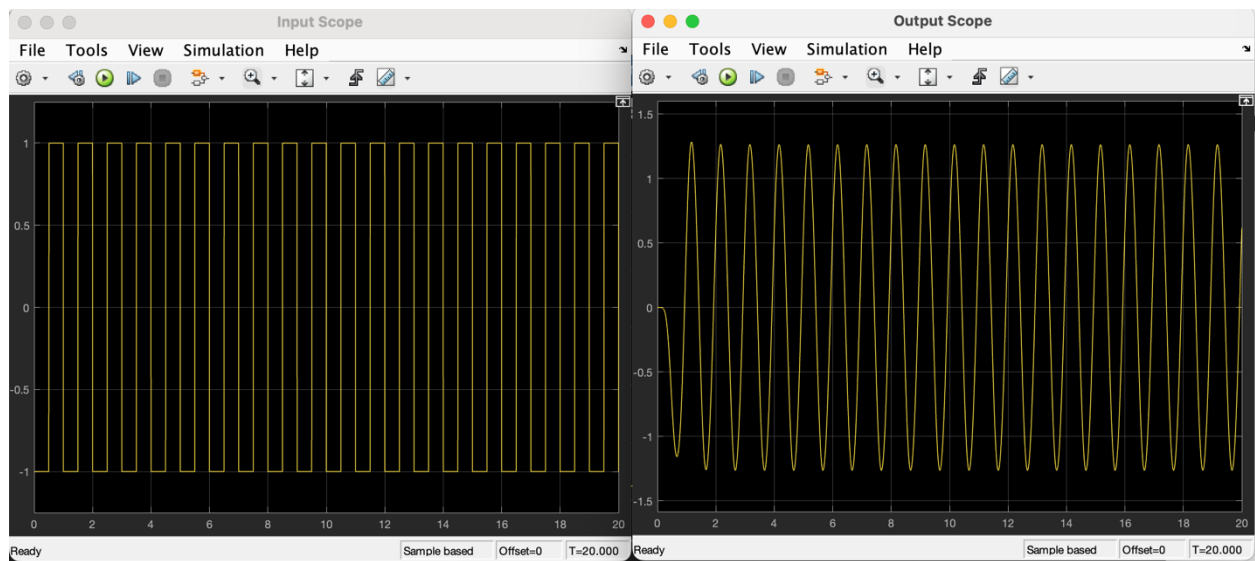
Part 2

Q 2.1:

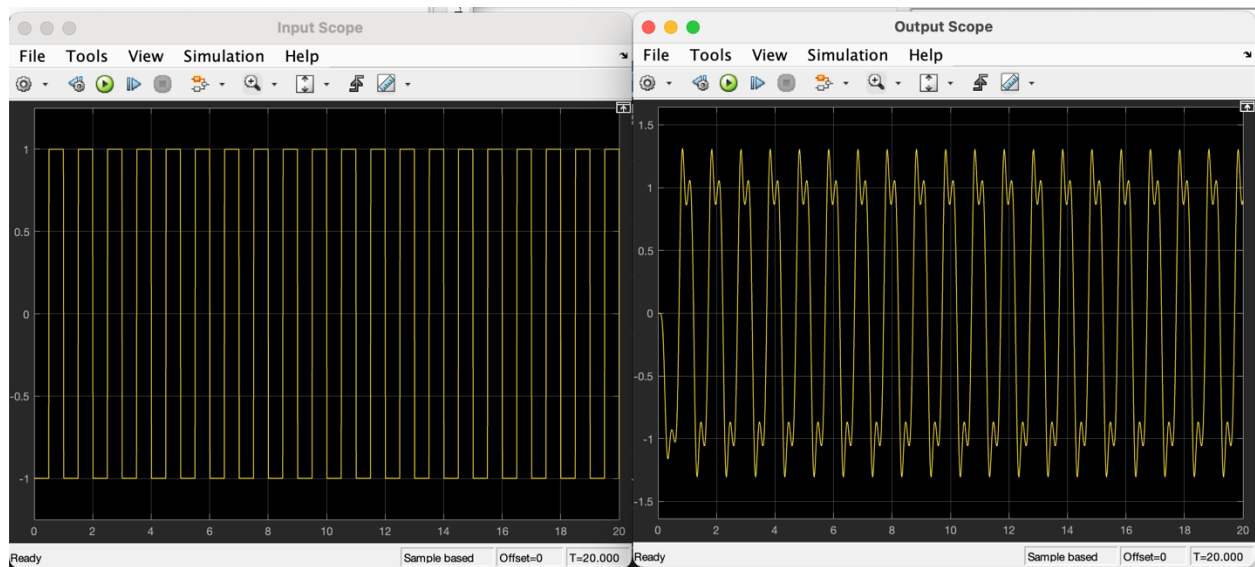
2.1.2



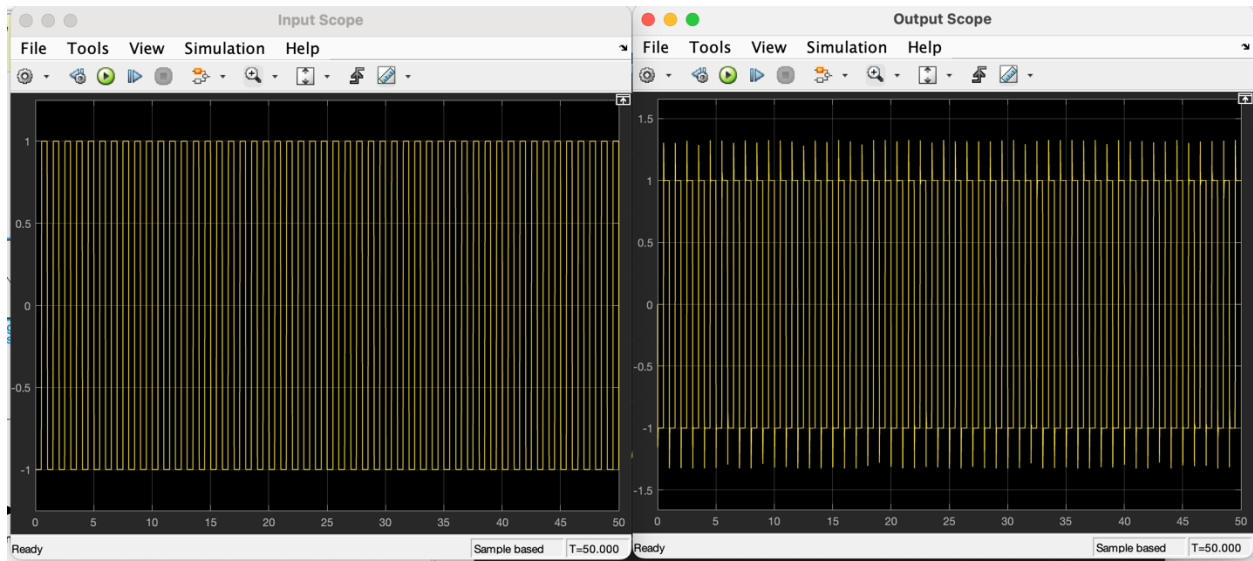
Cut-off frequency 1 Hz



Cut-off frequency 2 Hz



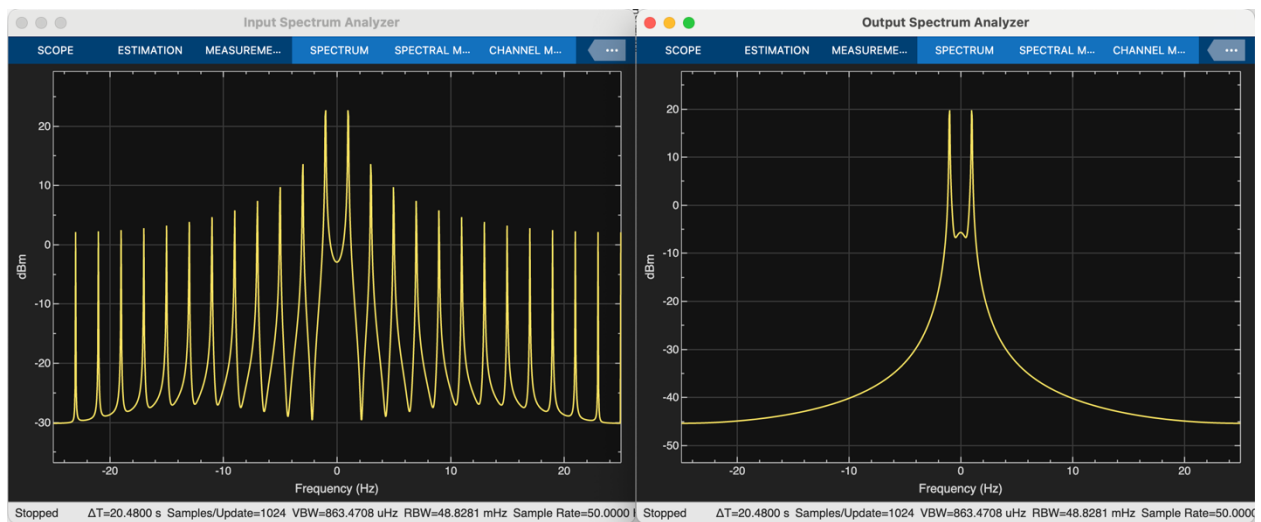
Cut-off frequency 4 Hz



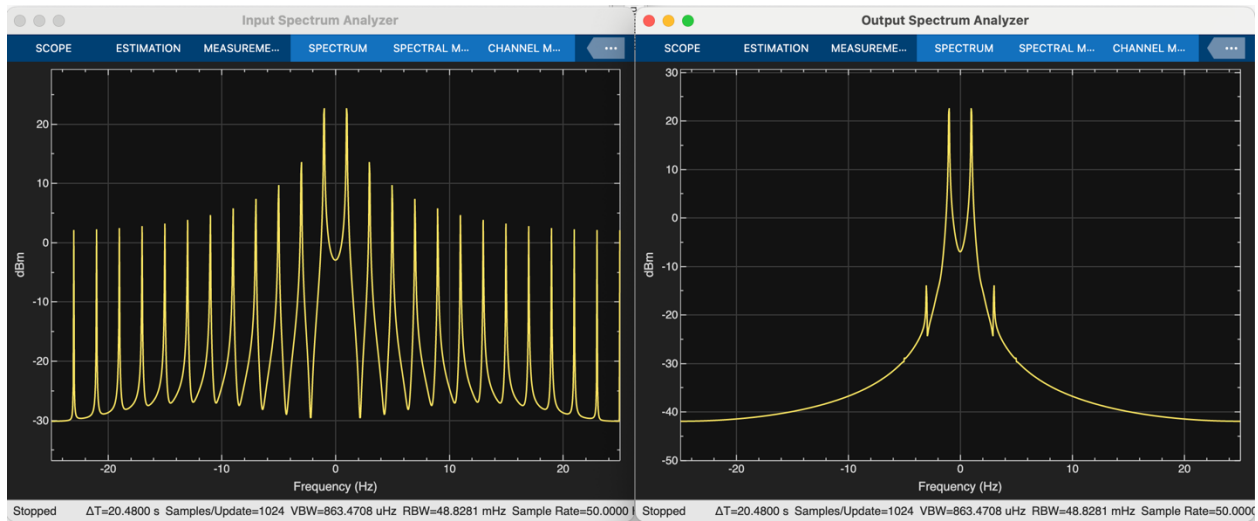
Cut-off frequency 100 Hz

2.1.4

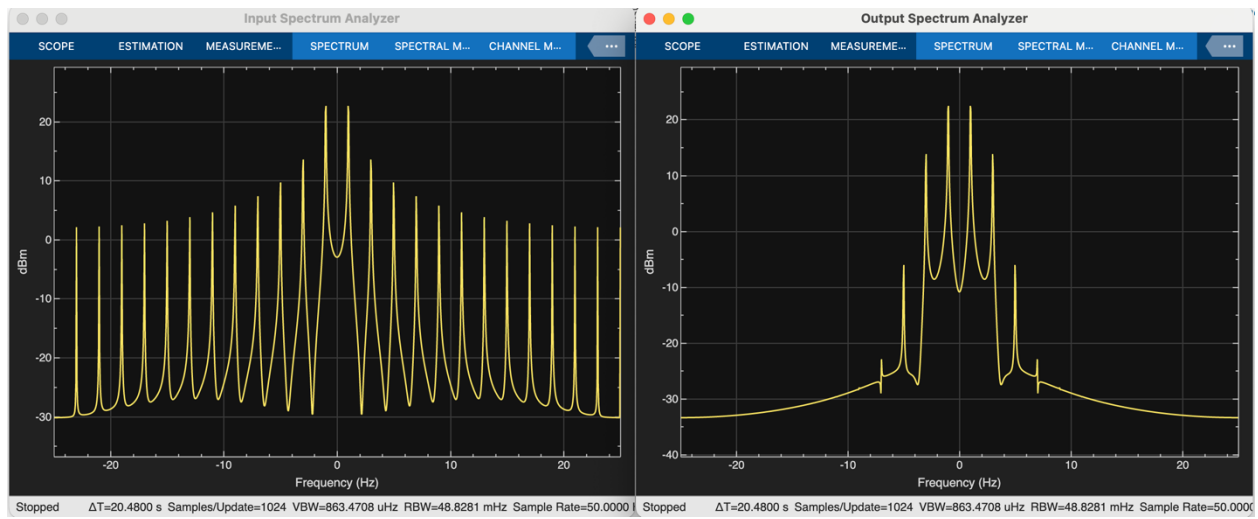
The frequency is varied from 2Hz to 100 Hz and the observations are displayed in the graphs below:



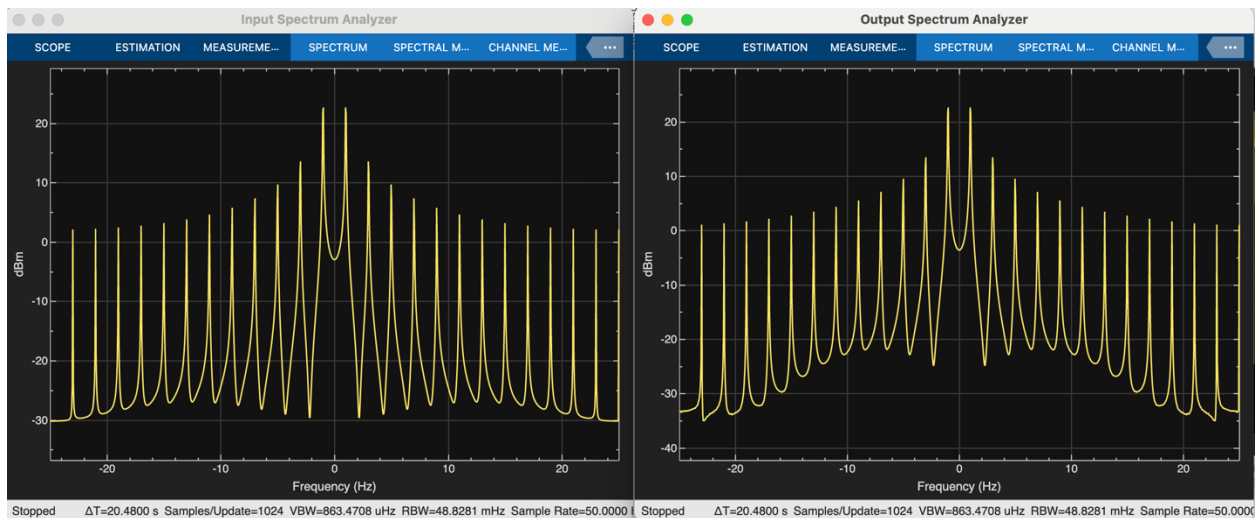
Cut-off frequency 1 Hz



Cut-off frequency 2 Hz



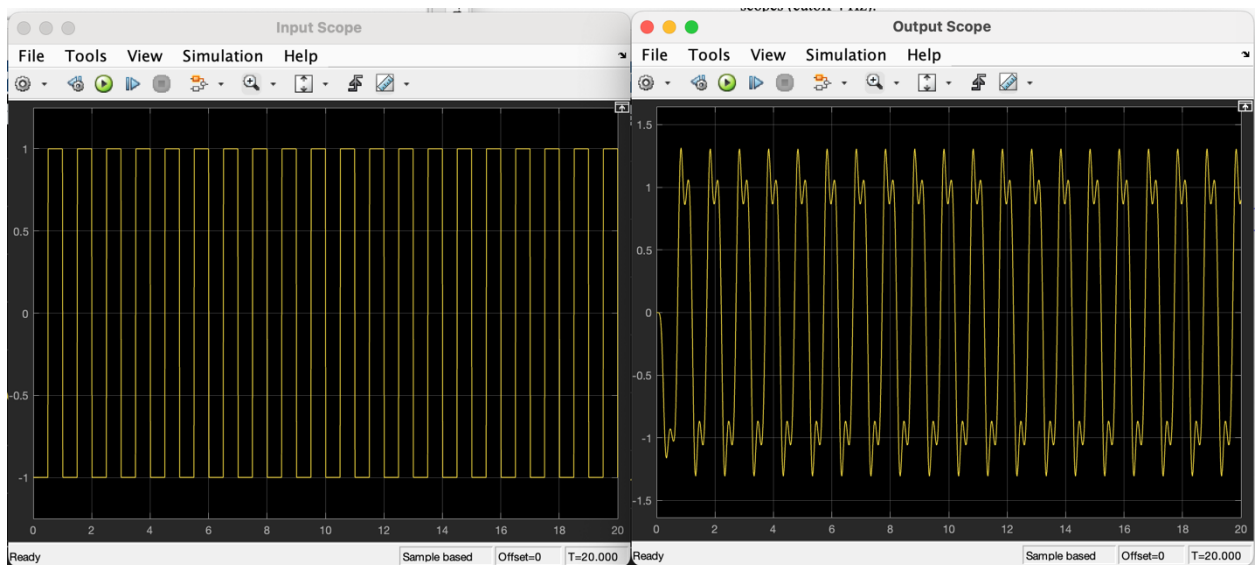
Cut-off frequency 4 Hz

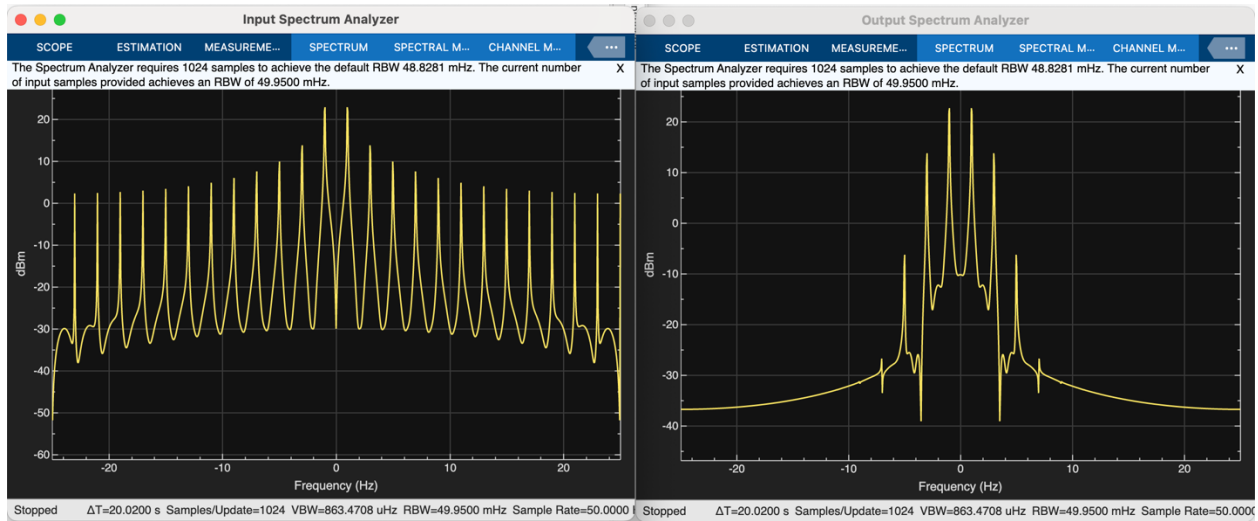


Cut-off frequency 100 Hz

Q 2.2:

Plot in one sheet four graphs corresponding to the input and output waveforms displayed by the time and frequency scopes (cut-off 4 Hz).





Q 2.3:

$$\begin{aligned}
 h(t) &= 1 \Pi t - 0.75 \ 0.5 \\
 &= 0.5 \text{Sa}(0.5\pi f) e^{-j2\pi f 0.75} \\
 &= 0.5 \text{Sa}(0.5\pi f) e^{-j1.5\pi f} \\
 C(n) &= f(0)H(nf(0)) \\
 &0.34 \sin(0.34\pi n) \ 0.34\pi n \\
 C1 &= 0.302 \\
 C2 &= 0.094 \\
 C3 &= -0.062 \\
 C4 &= -0.076
 \end{aligned}$$

Theoretical (dB)	Graphical dB
-16.34	-16.82
4.36	4.59
2.74	2.86

Table 1 – Theoretical and experimental values of amplitude spectrum of input signal

Q 2.4:

Frequency	H(f) Theoretical (dB)	H(f) Graphical dB
H(3)	-0.28343	-0.28547
H(4)	-4.3782	-4.3795
H(5)	-15.9325	-16.1834

Table 2 – Theoretical and experimental values of H(f)

From Table 2 it can be deduced that the values obtained experimentally and by using the equation in Q1.4 are consistent